ASTRA: cosmic-web environment splitting as a cosmological probe

Quantile-split clustering results: math, validation, and Fisher forecasts

ASTRA clustering project

2026

Outline

Why go beyond the two-point correlation function?

- The standard cosmological probe is the galaxy two-point correlation function (2PCF) $\xi(s)$ — a single number per separation bin, averaged over *all* environments.
- But galaxies live in very different places: deep voids, sheets, filaments, dense knots. These environments respond differently to:
  - **cosmology** (growth of structure, σ_8, ω_{cdm}, $\ldots$)
  - **galaxy–halo physics** (the HOD: how galaxies populate halos)
- **The bet:** if we split clustering *by local environment*, we get extra handles that respond differently to these two effects — breaking degeneracies that the plain 2PCF cannot.

This talk

Can we actually classify environment cheaply and robustly, measure real physics with it, and turn that into tighter cosmological constraints? We answer this with the ASTRA method on AbacusSummit simulations.

ASTRA: classifying the cosmic web

Idea: build a Delaunay tessellation over data + random points; a point's local density is read off from how many of its tessellation neighbours are data vs. random.

$$r = \frac{n_{\mathrm{data}} - n_{\mathrm{rand}}}{n_{\mathrm{data}} + n_{\mathrm{rand}}} \in [-1, 1]$$

Type	r range	meaning
Void	$-1 \leq r \leq -0.9$	far fewer data neighbours
Sheet	$-0.9 < r \leq 0$	mildly underdense
Filament	$0 < r \leq 0.9$	mildly overdense
Knot	$0.9 < r \leq 1$	almost all neighbours are data

Galaxies (and, independently, randoms) are then split into **quantiles** of r: Q1 = most underdense → Q4 = most overdense. Quantiles use *equal counts*; the physical void/sheet/filament/knot classes above do not (voids and knots turn out to be tiny populations — more on this later).

From classification to a data vector: the ASTRA "legs"

Once every data and random point has an r (hence a quantile label $Q1$–$Q4$), we measure the redshift-space multipoles $\xi_\ell(s)$ ($\ell = 0$ monopole, $\ell = 2$ quadrupole) of every combination:

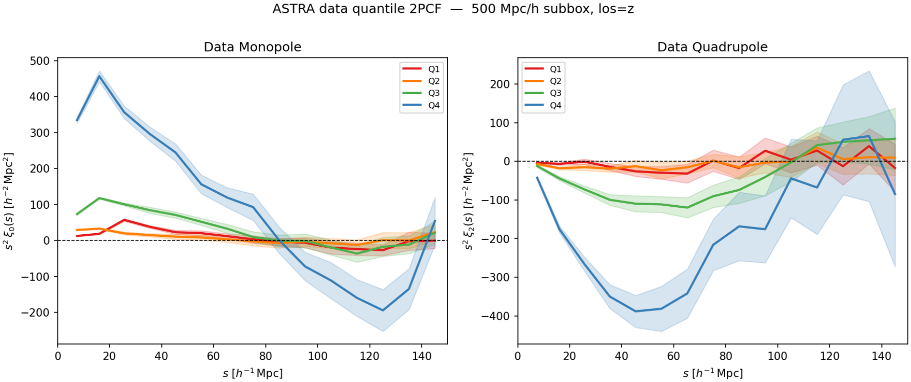
- **full sample auto** — the plain 2PCF baseline
- **data quantile autos** $dQ1..dQ4$
- **random quantile autos** $rQ1..rQ4$ (yes — ASTRA randoms trace the same cosmic web, so their quantiles cluster too!)
- **full** $\times$ **data-quantile crosses** $xdQ1..xdQ4$
- **full** $\times$ **random-quantile crosses** $xrQ1..xrQ4$

That is 17 stems $\times \{\ell = 0, 2\} \times$ 15 separation bins per simulated box: a rich, structured data vector built entirely out of standard 2PCF machinery, applied per environment.

Measurement pipeline

- AbacusSummit HOD galaxy catalogs ($2\,h^{-1}\mathrm{Mpc}[\mathrm{Gpc}]$ boxes, $\sim 4 \times 10^6$ galaxies), redshift-space positions (RSD along the line of sight) and Alcock–Paczynski rescaled into a common fiducial frame.
- **Two random catalogs, kept strictly separate:**
  - *ASTRA randoms* ($1\times$ data): enter the Delaunay tessellation, get their own *r* and quantile label, regenerated every iteration.
  - *Geometry randoms* ($5\times$ data, fixed): the plain Landy–Szalay reference, never touch the classification.
- Multipoles measured with the Landy–Szalay estimator, $\hat{\xi} = (DD - 2DR + RR)/RR$, using $\texttt{pycorr}/\texttt{Corrfunc}$.
- Everything is repeated over several independent ASTRA-random realisations and averaged, so the classification noise itself is controlled.

What does ASTRA actually measure?



Bottom line

A clean monotonic ordering Q1 $<$ Q2 $<$ Q3 $<$ Q4 in clustering amplitude — knots cluster far more strongly than the general sample, voids barely at all: the expected cosmic-web signature, from a Delaunay triangulation and a standard 2PCF.

A sanity check: why must $\xi(s)$ cross zero?

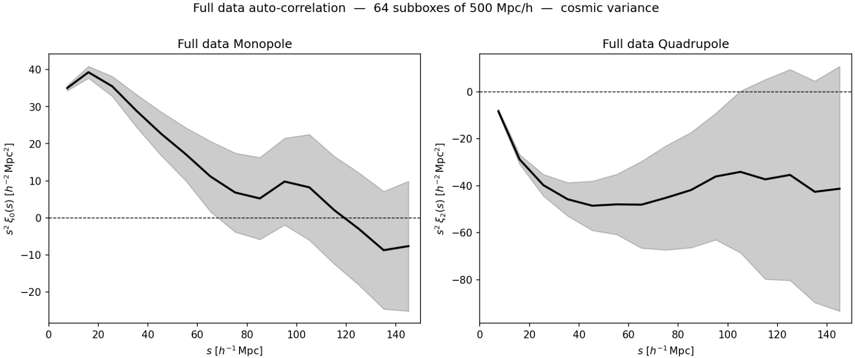
$$\xi(r) = \frac{1}{2\pi^2}\int_0^\infty dk\, k^2 P(k)\, j_0(kr), \qquad P(k\!=\!0) = \int \xi(r)\, d^3r.$$

- For a primordial spectrum with $n_s > 0$, $P(k\to 0) \to 0$: the strong small-scale positive correlation must be compensated by a shallow negative tail. Every overdense region sits in a slightly underdense larger environment.
- The crossing scale tracks the horizon at matter–radiation equality (Prada et al. 2011) — a standard ruler, $s_0 \approx 125$–$130\, h^{-1}\mathrm{Mpc}$ in linear theory.
- At leading order, bias and Kaiser RSD only *rescale* ξ: $\xi_g = b^2\xi_m$. They cannot move the zero. **Every linearly biased tracer must cross zero at the same scale.**

Why this matters for ASTRA

If our environment-selected quantiles are behaving as genuine, linearly biased tracers of the same underlying field — not noise, not an artefact of the classification — they must all share this crossing.

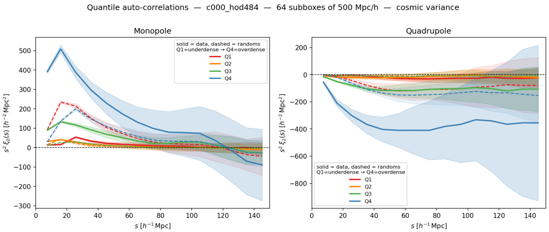
Validation I: the crossing is where it should be

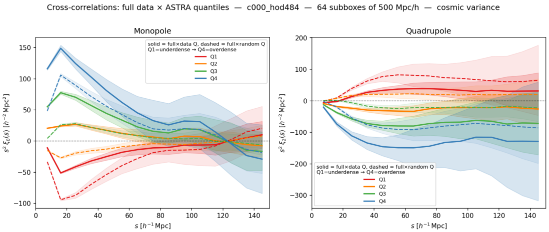


Bottom line

The full-sample monopole crosses zero at $s \simeq 123\,h^{-1}\mathrm{Mpc}$, matching the linear-theory equality-horizon prediction to within the expected integral-constraint shift of our finite subbox volume. The pipeline reproduces known physics before we ever split by environment.

Validation II: every ASTRA statistic shares the crossing





Bottom line

Despite wildly different amplitudes (knots cluster $\sim 10\times$ more strongly than voids), **every** data and random quantile auto- and cross-correlation crosses zero at the same $s_0 \simeq 117\text{–}123\, h^{-1}\mathrm{Mpc}$ (Q1's cross approaches zero from below, consistent with anti-biased voids). This is a non-trivial, passed consistency test: ASTRA quantiles behave as genuine linearly biased tracers of the same cosmic density field — real physics, not classification noise.

The Fisher-matrix formalism

For a Gaussian data vector $\boldsymbol{\xi}$ with covariance C,

$$F_{\alpha\beta} = \frac{\partial \boldsymbol{\xi}}{\partial \theta_\alpha}^{\top} \mathsf{C}^{-1} \frac{\partial \boldsymbol{\xi}}{\partial \theta_\beta}, \qquad \sigma(\theta_\alpha) = \left[(F^{-1})_{\alpha\alpha}\right]^{1/2}.$$

- Quality is set entirely by two ingredients: the **derivatives** $\partial\xi/\partial\theta$ and the **covariance** C.
- The inverse covariance estimated from N samples over p bins is biased; we apply the Hartlap correction $(N-p-2)/(N-1)$ throughout. This is what caps how long a data vector can be for a fixed number of covariance samples — a recurring theme below.
- Target parameters: $\{\omega_b, \omega_{cdm}, n_s, \sigma_8\}$.

The simulation grid

Nine AbacusSummit boxes ($2\,h^{-1}\mathrm{Mpc}[\mathrm{Gpc}]$, phase `ph000`), fiducial plus four $\pm$ pairs of the linear derivative grid, each populated with an LRG-like HOD catalog ($\sim 4 \times 10^6$ galaxies):

Cosmology	Variation	HOD catalog
c000	fiducial (Planck 2018)	hod484
c100/c101	$\pm 2\%$ in $\ln \omega_b$	hod179/hod152
c102/c103	$\pm 3.3\%$ in $\ln \omega_{cdm}$	hod556/hod861
c104/c105	± 0.01 in n_s	hod498/hod589
c112/c113	$\pm 2\%$ in σ_8	hod507/hod483

Cosmology derivatives come from central differences within each $\pm$ pair; the covariance comes from $500\,h^{-1}\mathrm{Mpc}$ subboxes of the fiducial box.

A hard lesson: HOD contamination

The core problem

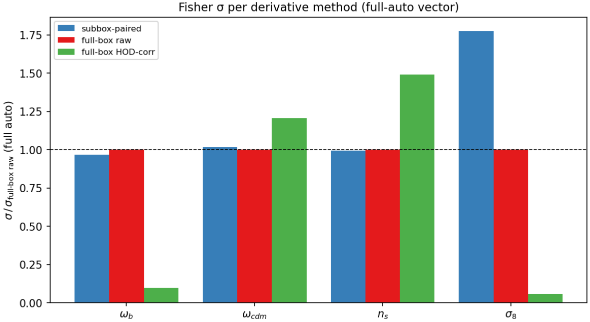
The available HOD catalogs are **not matched across cosmologies**: each cosmology carries its own random draw from the 12-parameter HOD prior. A naive $\pm$-pair derivative mixes the cosmology signal with a spurious galaxy–halo difference.

- Measured directly, the c100 $-$ c101 difference is a scale-dependent -6% to $+9\%$ tilt in ξ_0 — against a $\sim 1\%$ physical ω_b signal.
- The σ_8 pair difference loses $\sim 89\%$ of its amplitude to a cancelling HOD mismatch. One-sided or naive paired derivatives are unusable for σ_8.

Removing this contamination is the thread connecting every method that follows.

Fix 1 (Tier 0): full-box, phase-matched derivatives

Differencing the *whole* $2\,h^{-1}\mathrm{Mpc}[\mathrm{Gpc}]$ box (not noisy $500\,h^{-1}\mathrm{Mpc}$ subboxes) between the $\pm$ pair, with a noise model from the ASTRA-iteration scatter.

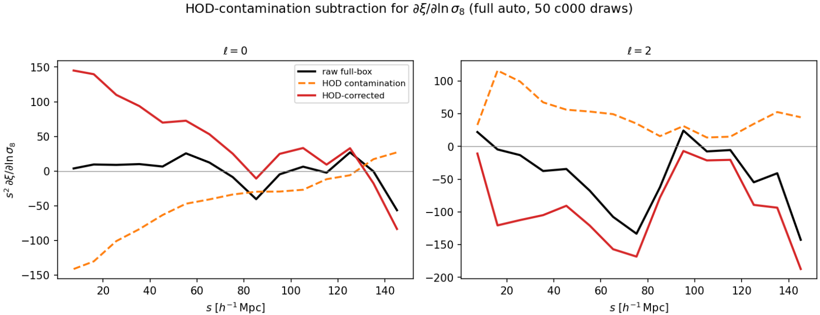


Bottom line

For the three already-clean parameters, subbox and full-box derivatives agree to $\approx$3%. For σ_8, removing derivative *noise* tightens $\sigma(\ln\sigma_8)$ by $\sim$1.8$\times$ — but this fixes noise, not the HOD contamination *bias* above.

Fix 2 (Tier 1): explicit HOD-contamination subtraction

Measure $\partial\xi/\partial\theta_{\mathrm{HOD}}$ by regressing ξ on 12 standardised HOD parameters across $\sim$50 extra fiducial-cosmology HOD draws, then subtract the modelled contamination from each pair's derivative.



Bottom line

Linear theory predicts $\partial\xi/\partial \ln\sigma_8 = 2\xi$. The raw derivative was only 11% of that; the corrected derivative recovers 136% — back in the right regime. The contamination (orange) rivals or exceeds the raw signal (black) everywhere: these derivatives were unusable before this fix.

Covariance: pooling across cosmologies

- The 64 fiducial subboxes alone cap the data vector at $p < 64$ bins — adding the quadrupole to the quantile vectors pushes Hartlap negative.
- Since the covariance varies far more slowly with cosmology than the mean does, we pool mean-subtracted subboxes across all **nine** cosmologies:

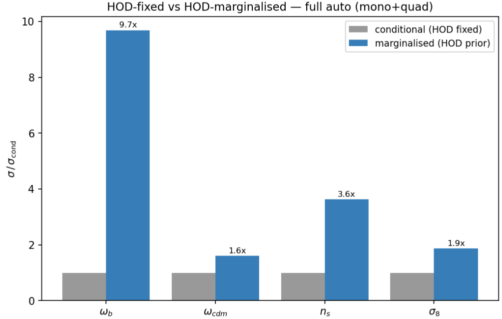
$$\mathrm{C}_{\mathrm{sub}} = \frac{1}{N - n_c} \sum_{c=1}^{9} \sum_{i=1}^{64} \left(\boldsymbol{\xi}_{ci} - \bar{\boldsymbol{\xi}}_c\right)\left(\boldsymbol{\xi}_{ci} - \bar{\boldsymbol{\xi}}_c\right)^{\top}, \qquad N = 9 \times 64 = 576.$$

Bottom line

With $N = 576$ the Hartlap factor stays $\gtrsim 0.75$ even at $p = 144$ bins — this single change is what lets the quadrupole and the random-quantile legs into the data vector at all.

Fixing the HOD exactly is optimistic

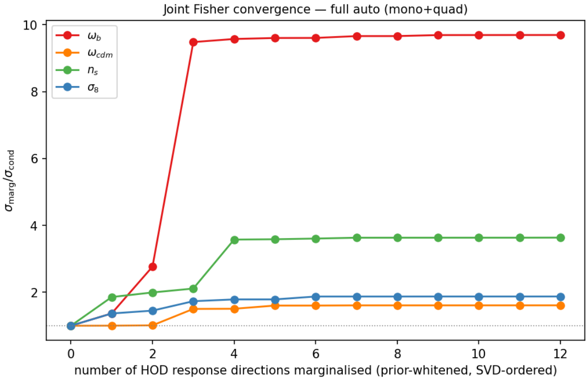
We build one Fisher over 4 cosmology + 12 HOD parameters and marginalise the HOD under a Gaussian yuan23 prior, for the plain full-sample 2PCF (mono+quad).



Bottom line

Marginalising over the 12 HOD nuisance parameters degrades every cosmological error — worst for ω_b (**9.7**$\times$), mildest for ω_{cdm} (1.6$\times$). This is the honest starting point: exactly what a smarter data vector needs to claw back.

The HOD response is low-rank

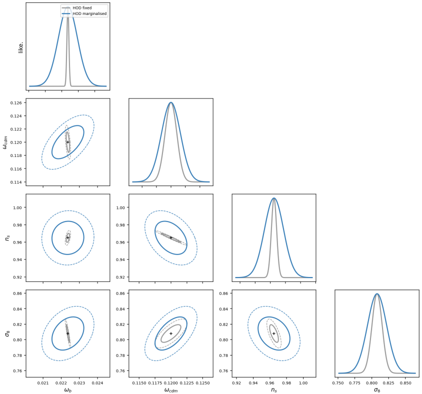


Bottom line

Every parameter's marginalised error plateaus after $\sim$4 marginalised HOD directions and stays flat through all 12 — the correction is robust to the prior tail, not an artefact of how the HOD prior was modelled.

The same story, in cosmological parameter space

Joint Fisher (68% solid / 95% dashed): HOD-fixed (grey) vs HOD-marginalised (blue) — full auto (mono+quad)



Bottom line

Grey = HOD fixed, blue = HOD marginalised, full-sample auto alone. The blue ellipses are dramatically larger *and* tilted differently: this is the ground the ASTRA environment split needs to win back.

Headline: ASTRA quantile splits win back the ground

Data vector (ℓ's)	p	Hartlap	$\sigma(\omega_b)$	$\sigma(\omega_{cdm})$	$\sigma(n_s)$
full auto (mono+quad)	30	0.95	2.18	9.31	52.3
data Q autos (mono)	32	0.94	1.94	7.53	29.9
data Q autos (mono+quad)	64	0.89	1.46	5.51	18.1
full+data Q (mono+quad)	80	0.86	**1.14**	**4.47**	**15.5**

(units 10^{-4}; HOD-marginalised, pooled 576-sample covariance)

Bottom line

Adding the ASTRA environment quantiles *and* the quadrupole to the plain 2PCF tightens every parameter by **1.9**$\times$ **(**ω_b**), 2.1**$\times$ **(**ω_{cdm}**), 3.4**$\times$ **(**n_s**)** — *after* full HOD marginalisation. The earlier finding that "the quadrupole adds nothing" was a Hartlap-budget artefact of the smaller 64-subbox covariance; with the pooled covariance both the environment split and the quadrupole are real, independent information.

Sharpening the derivatives: a global response model

Instead of patching contamination pair-by-pair, fit *one* linear response across ∼50 maximin HOD draws *per* cosmology:

$$\boldsymbol{\xi}(\boldsymbol{\theta}) \approx \boldsymbol{\xi}_0 + \sum_p a_p\, \theta_{\mathrm{cosmo},p} + \sum_q b_q\, \theta_{\mathrm{HOD},q}.$$

The HOD term absorbs the cross-cosmology mismatch *by construction*: a_p is HOD-clean without any subtraction step, and cosmic variance cancels because every run shares phase `ph000`.

Data vector	$\sigma(\omega_b)$	$\sigma(\omega_{cdm})$	$\sigma(n_s)$	$\sigma(\sigma_8)$
full auto (mono+quad)	2.12	9.74	52.6	81.5
full+data+rand Q (mono+quad)	**0.69**	**2.72**	**9.76**	**36.5**

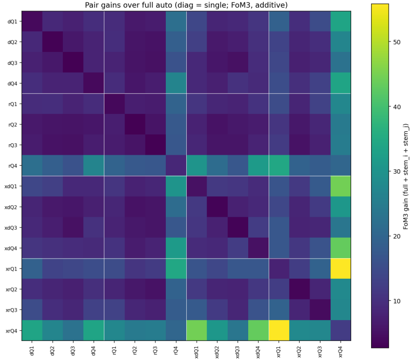
(units 10^{-4})

Bottom line

Adding the *random*-quantile legs on top tightens further, by 3–5× over the plain 2PCF on every parameter: random quantiles respond to cosmology and HOD differently from the galaxies, breaking degeneracies the galaxy clustering alone cannot.

Searching systematically for the best combination

Sweep all 16 single stems and all 120 pairs, scored by the HOD-marginalised figure of merit $\mathrm{FoM}_3 = [\det \mathrm{Cov}_3]^{-1/2}$.

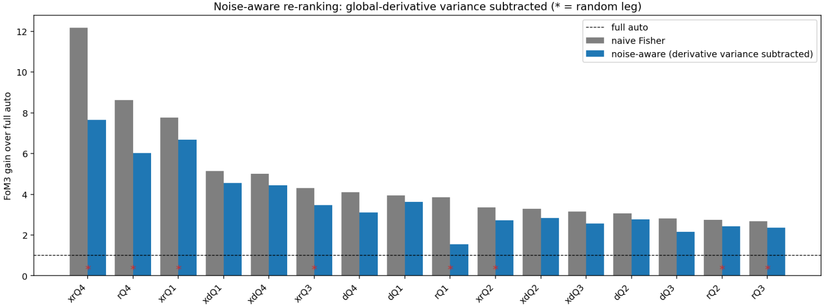


Bottom line

One cell dominates the entire 16×16 grid: **full**$+$*xrQ*4 $+$ *xrQ*1 (full sample crossed with the knot- and void-environment *randoms*) — the brightest square on the map, outscoring every other combination.

Warning sign: is this just noise?

The naive ranking is dominated by vectors with a **random**-quantile leg (gains up to 55$\times$) — surprising, since random quadrupoles are known to be noise-limited. Test 1: subtract the derivative-noise bias $\mathrm{Tr}(C^{-1}\mathrm{Cov}(\delta))$ from every candidate.



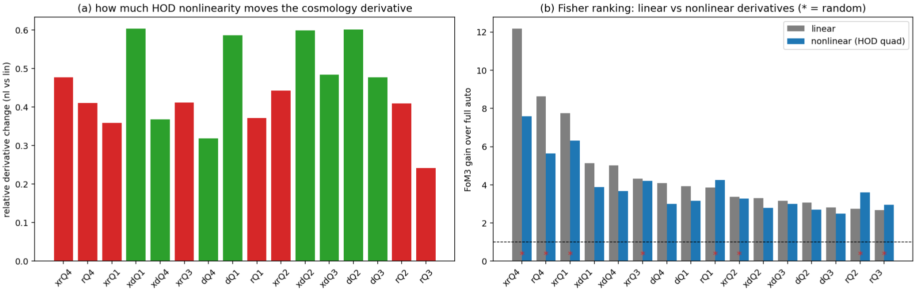
Bottom line

The full auto has zero correction (a clean check); the random legs ($\star$) deflate only $\sim$25–40% and **stay on top** — their environment-classified *monopoles* carry real signal, not noise.

Warning sign: is this just HOD nonlinearity?

Test 2: the global response is linear in the HOD; refit with quadratic HOD terms to see whether nonlinearity preferentially inflates the random-leg derivatives.

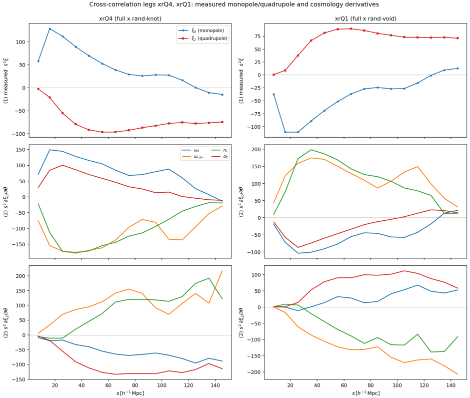
Nonlinear-response test of the random-leg cosmology derivatives



Bottom line

(a) Data legs (green) move *as much or more* than random legs (red) when nonlinearity is added. (b) Ranking unchanged — random legs still on top. Nonlinearity does not single them out either.

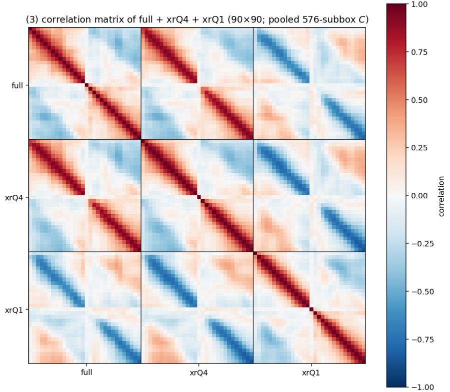
Why $xrQ4$ and $xrQ1$ are mirror images



Bottom line

$xrQ4$ (full$\times$knot-random, left) and $xrQ1$ (full$\times$void-random, right) are near mirror images: opposite measured signal ($\xi_0 = +0.50$ vs. -0.43 at $s \sim 15\,h^{-1}\mathrm{Mpc}$) and opposite cosmology response ($\partial\xi_0/\partial \ln \omega_{cdm} = -0.60$ vs. $+0.48$).

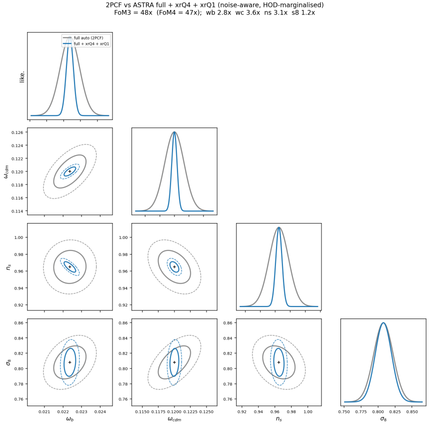
Mirror images, anti-correlated



Bottom line

The full–$xrQ4$ block is positively correlated (red); the full–$xrQ1$ and $xrQ4$–$xrQ1$ blocks are *negative* (blue). Opposite signal, opposite response, *and* anti-correlated noise: exactly the recipe for strong Fisher leverage from two extra statistics.

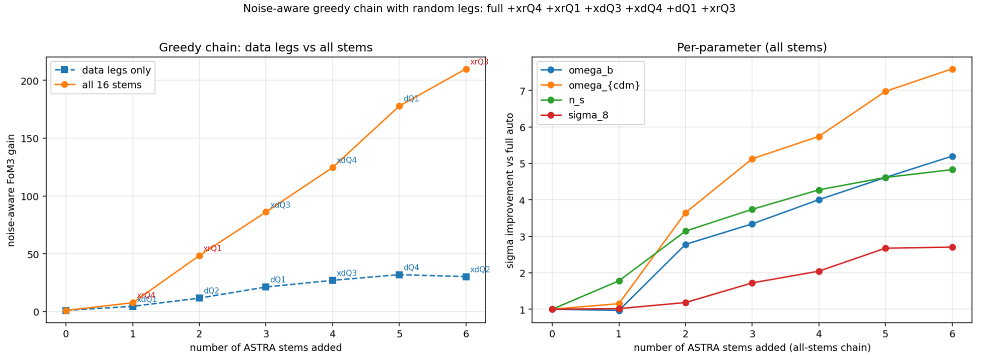
The payoff, in cosmological parameter space



Bottom line

Adding *only two* statistics — the void and knot random crosses — to the plain 2PCF already gives $\mathrm{FoM}_3 = 48\times$ ($\omega_b\, 2.8\times$, $\omega_{cdm}\, 3.6\times$, $n_s\, 3.1\times$): **more than the entire five-stem data-quantile vector**. Ellipses are smaller *and* rotated.

Assembling the full optimal vector

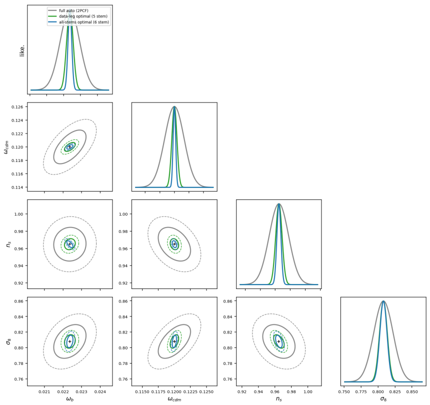


Bottom line

Greedy selection over all 16 stems picks the void/knot random crosses *first* ($1 \to 48\times$); three more data legs push the 6-statistic vector to $\mathrm{FoM}_3 \simeq 210\times$ — $\sigma(\omega_b)\,5.2\times$, $\sigma(\omega_{cdm})\,7.6\times$, $\sigma(n_s)\,4.8\times$, $\sigma(\sigma_8)\,2.7\times$ tighter than the plain 2PCF.

The forecast, side by side

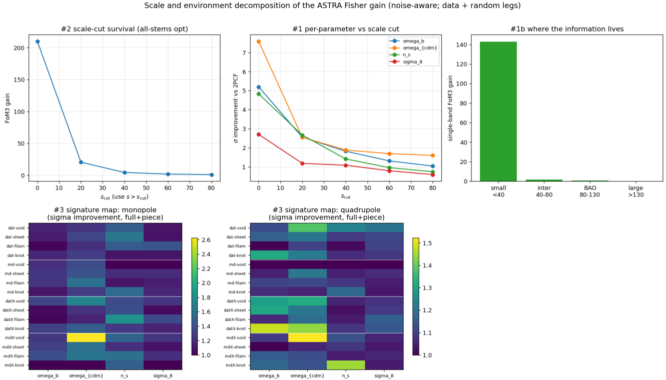
Updated Fisher forecast (noise-aware, HOD-marginalised): 2PCF vs ASTRA optimal vectors



Bottom line

Grey = plain 2PCF, green = the conservative data-only optimal vector (32×), blue = the all-stems ceiling led by the void/knot random crosses (210×). Both shrink *and rotate* the ellipses relative to the 2PCF — genuine degeneracy-breaking, not just a smaller error bar.

Where the gain lives: it's a small-scale phenomenon



Bottom line

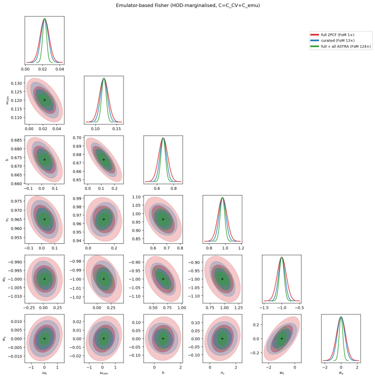
The 210× collapses to 20.6× at $s > 20\,h^{-1}\mathrm{Mpc}$, 4.4× at $s > 40\,h^{-1}\mathrm{Mpc}$: ASTRA splits by *local density*, and at BAO/large scales every quantile shares the same zero crossing (recall the validation!), so splitting adds nothing there. **Two honest headlines:** $\sim$1.5–1.9× with a conservative $s > 40\,h^{-1}\mathrm{Mpc}$ cut, or several-fold with a trusted small-scale model.

From forecast to measurement

The $\sim$200$\times$ headline needs *simulation-based* small-scale modelling (an emulator), not perturbation theory. This path is already built and validated:

- A neural emulator $\mu(\boldsymbol{\theta})$ trained across the AbacusSummit emulator grid gives HOD-clean derivatives at fixed HOD, for free, for all 8 $w_0 w_a$CDM+ parameters.
- Realistic covariance $C = C_{\rm CV} + C_{\rm emu}$ (cosmic variance *plus* emulator error), not the optimistic $C_{\rm CV}$ alone.
- The **curated vector** the emulator campaign settled on is exactly full 2PCF + the void-random cross monopole — the same $xrQ1$ leg that led the greedy chain above, now re-derived independently by the emulator pipeline.

The void-random leg strikes again, with a realistic error budget



Bottom line

Even with the realistic $C_{\rm CV} + C_{\rm emu}$ covariance (cosmic variance *plus* measured emulator error) over all 8 $w_0 w_a$CDM+ parameters: the curated vector (2PCF + void-random cross) gives FoM 14×; full ASTRA gives 150× — again far exceeding the ×1.1–2.2 per-parameter gain, the degeneracy-breaking signature. Validated end-to-end against an MCMC.

The key strategic insight

The value-add is emulator-gated, not information-limited

ASTRA (and any environment or marked statistic) carries real cosmological information — the Fisher gains above are real. But that information is only *realisable in inference* once the emulator error on the environment legs falls below cosmic variance.

- This reframes the open question from *"does the statistic help?"* (yes, robustly, by every test in this talk) to *"can we emulate it accurately enough?"* — a concrete, quantitative, and *solvable* target (denser cosmology coverage, the dominant lever).
- It is a general lesson for any simulation-based-inference programme built on marked or environment-split statistics, not just ASTRA.

Bottom line: why ASTRA has real potential for cosmology

1. **The classification is physical, not a fitting artefact** — every ASTRA statistic obeys the same zero-crossing consistency test as the plain 2PCF.
2. **The information is real and survives scrutiny** — vindicated against derivative-noise and nonlinear-HOD-response robustness checks, and against a full joint cosmology+HOD marginalisation, not just fixed-HOD optimism.
3. **One leg does most of the work**: the full$\times$void-random cross ($xrQ1$), paired with its knot mirror image ($xrQ4$), turns just *two* extra statistics into $\mathrm{FoM}_3 = 48\times$ — more than an entire five-stem data-quantile vector — traced to a specific, interpretable mechanism (opposite signal, opposite response, anti-correlated noise) and a specific scale range ($s \lesssim 40\,h^{-1}\mathrm{Mpc}$). It resurfaces independently as the emulator campaign's own curated vector.
4. **The path to a real measurement, not just a forecast, already exists** and is validated (emulator + MCMC), with a clear, quantitative target for what unlocks the rest of the gain.

Open challenges and outlook

- σ_8 remains the laggard throughout: amplitude-like, most degenerate with HOD bias, and least constrained by the current 9-cosmology grid.
- The random-leg gains are the box-only *ceiling*; in a real survey the randoms trace the selection window, so the conservative data-leg vector is the more directly transferable forecast until that window is forward-modelled.
- The covariance still rests on subboxes tiling a single simulation phase; an independent-phase mock suite (e.g. EZmocks) is the main residual weakness for an honest, unbiased real measurement.
- Next: finish the emulator campaign (denser cosmology coverage is the dominant lever), extend to w_0, w_a via the quadrupole, and validate on independent mocks.

Thank you